
EXAMPLE 9 An Inner Product on $C[a, b]$

Let $\mathbf{f} = f(x)$ and $\mathbf{g} = g(x)$ be two functions in $C[a, b]$ and define

$$\langle \mathbf{f}, \mathbf{g} \rangle = \int_a^b f(x)g(x)dx \quad (6)$$

This is well-defined since the functions in $C[a, b]$ are continuous. We shall show that this formula defines an inner product on $C[a, b]$ by verifying the four inner product axioms for functions $\mathbf{f} = f(x)$, $\mathbf{g} = g(x)$, and $\mathbf{s} = s(x)$ in $C[a, b]$:

1.
$$\langle \mathbf{f}, \mathbf{g} \rangle = \int_a^b f(x)g(x)dx = \int_a^b g(x)f(x)dx = \langle \mathbf{g}, \mathbf{f} \rangle$$

which proves that Axiom 1 holds.

2.
$$\begin{aligned} \langle \mathbf{f} + \mathbf{g}, \mathbf{s} \rangle &= \int_a^b (f(x) + g(x))s(x)dx \\ &= \int_a^b f(x)s(x)dx + \int_a^b g(x)s(x)dx \\ &= \langle \mathbf{f}, \mathbf{s} \rangle + \langle \mathbf{g}, \mathbf{s} \rangle \end{aligned}$$

which proves that Axiom 2 holds.

3.
$$\langle k\mathbf{f}, \mathbf{g} \rangle = \int_a^b kf(x)g(x)dx = k \int_a^b f(x)g(x)dx = k\langle \mathbf{f}, \mathbf{g} \rangle$$

which proves that Axiom 3 holds.

4. If $\mathbf{f} = f(x)$ is any function in $C[a, b]$, then $f^2(x) \geq 0$ for all x in $[a, b]$; therefore,

$$\langle \mathbf{f}, \mathbf{f} \rangle = \int_a^b f^2(x) dx \geq 0$$

Further, because $f^2(x) \geq 0$ and $\mathbf{f} = f(x)$ is continuous on $[a, b]$, it follows that $\int_a^b f^2(x) dx = 0$ if and only if $f(x) = 0$ for all x in $[a, b]$. Therefore, we have $\langle \mathbf{f}, \mathbf{f} \rangle = \int_a^b f^2(x) dx = 0$ if and only if $\mathbf{f} = \mathbf{0}$. This proves that Axiom 4 holds. ◆

EXAMPLE 10 Norm of a Vector in $C[a, b]$

If $C[a, b]$ has the inner product defined in the preceding example, then the norm of a function $\mathbf{f} = f(x)$ relative to this inner product is

$$\|\mathbf{f}\| = \langle \mathbf{f}, \mathbf{f} \rangle^{1/2} = \sqrt{\int_a^b f^2(x) dx} \quad (7)$$

and the unit sphere in this space consists of all functions f in $C[a, b]$ that satisfy the equation $\|\mathbf{f}\| = 1$, which on squaring yields

$$\int_a^b f^2(x) dx = 1$$

17. (For Readers Who Have Studied Calculus)

Let the vector space P_2 have the inner product

$$\langle \mathbf{p}, \mathbf{q} \rangle = \int_{-1}^1 p(x)q(x)dx$$

(a) Find $\|\mathbf{p}\|$ for $\mathbf{p} = 1$, $\mathbf{p} = x$, $\mathbf{p} = x^2$.

(b) Find $d(\mathbf{p}, \mathbf{q})$ if $\mathbf{p} = 1$ and $\mathbf{q} = x$.

② Find $\|p\|$ for $p=1$, $p=x$ and $p=x^2$.

If $p=1$, $\langle p, p \rangle = \int_{-1}^1 p^2 dx$
 $= \int_{-1}^1 1 dx = \sqrt{x} \Big|_{-1}^1 = \sqrt{2}$

If $p=x$, $\langle p, p \rangle = \int_{-1}^1 x^2 dx = \sqrt{\left(\frac{x^3}{3}\right)} \Big|_{-1}^1 = \sqrt{\frac{1}{3} + \frac{1}{3}} = \sqrt{\frac{2}{3}} = \frac{1}{3}\sqrt{6}$

If $p=x^2$, $\langle p, p \rangle = \int_{-1}^1 x^4 dx = \sqrt{\left(\frac{x^5}{5}\right)} \Big|_{-1}^1 = \sqrt{\frac{1}{5} + \frac{1}{5}} = \sqrt{\frac{2}{5}} = \frac{1}{5}\sqrt{10}$

⑥ $d(p, q)$ if $p=1$ and $q=x$.

$$d(p, q) = \langle p - q, p - q \rangle^{\frac{1}{2}} = \sqrt{\int_{-1}^1 (1-x)^2 dx}$$

$$= \sqrt{\int_{-1}^1 (1 + x^2 - 2x) dx} = \sqrt{\left[x + \frac{x^3}{3} - \frac{2x^2}{2}\right]_{-1}^1}$$

$$= \sqrt{\left(1 + \frac{1}{3} - 1\right) - \left(-1 - \frac{1}{3} - 1\right)}$$

$$= \sqrt{2 + \frac{2}{3}} = \sqrt{\frac{8}{3}} = \frac{2}{3}\sqrt{6}$$

17. (For Readers Who Have Studied Calculus)

Let the vector space P_2 have the inner product

$$\langle \mathbf{p}, \mathbf{q} \rangle = \int_{-1}^1 p(x)q(x)dx$$

(a) Find $\|\mathbf{p}\|$ for $\mathbf{p} = 1$, $\mathbf{p} = x$, $\mathbf{p} = x^2$.

(b) Find $d(\mathbf{p}, \mathbf{q})$ if $\mathbf{p} = 1$ and $\mathbf{q} = x$.

17. (a) For instance, $\|x\| = \left(\int_{-1}^1 x^2 dx \right)^{1/2} = \left(\left[\frac{x^3}{3} \right]_{-1}^1 \right)^{1/2} = \left(\frac{2}{3} \right)^{1/2}$

(b) We have

$$\begin{aligned} d(\mathbf{p}, \mathbf{q}) &= \|\mathbf{p} - \mathbf{q}\| \\ &= \|1 - x\| \\ &= \left(\int_{-1}^1 (1 - x)^2 dx \right)^{1/2} \\ &= \left(\int_{-1}^1 (1 - 2x + x^2) dx \right)^{1/2} \\ &= \left(\left[x - x^2 + \frac{x^3}{3} \right]_{-1}^1 \right)^{1/2} \\ &= 2 \left(\frac{2}{3} \right)^{1/2} = \frac{2}{3} \sqrt{6} \end{aligned}$$

27. (For Readers Who Have Studied Calculus)

Use the inner product

$$\langle \mathbf{p}, \mathbf{q} \rangle = \int_{-1}^1 p(x)q(x)dx$$

to compute $\langle \mathbf{p}, \mathbf{q} \rangle$ for the vectors $\mathbf{p} = p(x)$ and $\mathbf{q} = q(x)$ in P_3 .

(a) $\mathbf{p} = 1 - x + x^2 + 5x^3$, $\mathbf{q} = x - 3x^2$

(a) $p = 1 - x + x^2 + 5x^3$ $q = x - 3x^2$

$$\begin{aligned} \langle p, q \rangle &= \int_{-1}^1 p(x)q(x) dx \\ &= \int_{-1}^1 (1 - x + x^2 + 5x^3)(x - 3x^2) dx \\ &= \int_{-1}^1 (x - x^2 + x^3 + 5x^4 - 3x^2 + 3x^3 - 3x^4 - 15x^5) dx \\ &= \int_{-1}^1 (x - 4x^2 + 4x^3 + 2x^4 - 15x^5) dx \\ &= \left[\frac{x^2}{2} - \frac{4x^3}{3} + \frac{4x^4}{4} + \frac{2x^5}{5} - \frac{15x^6}{6} \right]_{-1}^1 \\ &= -\frac{1}{2}[1+1] + \frac{2}{5}[1+1] \\ &= -\frac{8}{3} + \frac{4}{5} = \frac{-40+12}{15} = \underline{\underline{-\frac{28}{15}}} \end{aligned}$$

27. (For Readers Who Have Studied Calculus)

Use the inner product

$$\langle \mathbf{p}, \mathbf{q} \rangle = \int_{-1}^1 p(x)q(x)dx$$

to compute $\langle \mathbf{p}, \mathbf{q} \rangle$ for the vectors $\mathbf{p} = p(x)$ and $\mathbf{q} = q(x)$ in P_3 .

(b) $\mathbf{p} = x - 5x^3, \quad \mathbf{q} = 2 + 8x^2$

$$\begin{aligned} \langle \mathbf{p}, \mathbf{q} \rangle &= \int_{-1}^1 p(x)q(x)dx = \int_{-1}^1 (x - 5x^3)(2 + 8x^2)dx \\ &= \int_{-1}^1 (2x - 10x^3 + 8x^3 - 40x^5)dx \\ &= \int_{-1}^1 (2x - 2x^3 - 40x^5)dx \\ &= \left[x^2 - \frac{2}{4}x^4 - \frac{40}{6}x^6 \right]_{-1}^1 = \underline{\underline{0}} \end{aligned}$$

28. (For Readers Who Have Studied Calculus)

In each part, use the inner product

$$\langle \mathbf{f}, \mathbf{g} \rangle = \int_0^1 f(x)g(x)dx$$

to compute $\langle \mathbf{f}, \mathbf{g} \rangle$, for the vectors $\mathbf{f} = f(x)$ and $\mathbf{g} = g(x)$ in $C[0, 1]$.

(a) $\mathbf{f} = \cos 2\pi x, \quad \mathbf{g} = \sin 2\pi x$

(b) $\mathbf{f} = x, \quad \mathbf{g} = e^x$

(c) $\mathbf{f} = \tan \frac{\pi}{4}x, \quad \mathbf{g} = 1$

(a) $f = \cos(2\pi x), g = \sin(2\pi x)$
 $\langle f, g \rangle = \int_0^1 \frac{1}{2} \sin(4\pi x) dx$
 $= 0$

(b) $f = x, g = e^x$
 $= \int_0^1 x e^x dx$
 $= [x e^x]_0^1 - \int_0^1 e^x dx$
 $= 1 \cdot e^1 - [e^x]_0^1$
 $= e - (e - 1)$
 $= 1$

(c) $f = \tan(\frac{\pi x}{4}), g = 1$
 $\frac{2 \ln(2)}{\pi}$

Theorem 6.1.1

(Properties of Inner Products)

- If $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are vectors in a real inner product space, and k is any scalar, then:
 - $\langle \mathbf{0}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{0} \rangle = 0$
 - $\langle \mathbf{u}, \mathbf{v} + \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{u}, \mathbf{w} \rangle$
 - $\langle \mathbf{u}, k\mathbf{v} \rangle = k \langle \mathbf{u}, \mathbf{v} \rangle$
 - $\langle \mathbf{u} - \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle - \langle \mathbf{v}, \mathbf{w} \rangle$
 - $\langle \mathbf{u}, \mathbf{v} - \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle - \langle \mathbf{u}, \mathbf{w} \rangle$

Assignments

- Exercise-6.1: Examples 1 – 11 (Pages 444-453)
- Exercise-6.1: Problems 1– 28 (Pages 453-458)

Chapter Content

- Inner Products
- Angle and Orthogonality in Inner Product Spaces
- Orthonormal Bases; Gram-Schmidt Process;
- Best Approximation; Least Squares

Theorems 6.2.1 & 6.2.2

- Theorem 6.2.1 (Cauchy-Schwarz Inequality)

- If $\mathbf{u}$ and $\mathbf{v}$ are vectors in a real inner product space, then

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \leq ||\mathbf{u}|| \, ||\mathbf{v}||$$

- Theorem 6.2.2 (Properties of Length)

- If $\mathbf{u}$ and $\mathbf{v}$ are vectors in an inner product space V , and if k is any scalar, then :

- $||\mathbf{u}|| \geq 0$

- $||\mathbf{u}|| = 0$ if and only if $\mathbf{u} = \mathbf{0}$

- $||k\mathbf{u}|| = |k| \, ||\mathbf{u}||$

- $||\mathbf{u} + \mathbf{v}|| \leq ||\mathbf{u}|| + ||\mathbf{v}||$ (Triangle inequality)

12. In each part, verify that the Cauchy-Schwarz inequality holds for the given vectors using the Euclidean inner product.

(a) $\mathbf{u} = (3, 2), \mathbf{v} = (4, -1)$

a) $\mathbf{u} = (3, 2), \mathbf{v} = (4, -1)$

$$\|\mathbf{u}\| = \sqrt{9+4} = \sqrt{13} \quad \|\mathbf{v}\| = \sqrt{16+1} = \sqrt{17}$$

$$\|\mathbf{u}\| \|\mathbf{v}\| = 14.866$$

$$\langle \mathbf{u}, \mathbf{v} \rangle = 3 \times 4 - 2 = 12 - 2 = 10 \quad \therefore |\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|$$

⑥ $\mathbf{u} = (-3, 1, 0), \mathbf{v} = (2, -1, 3)$

$$\|\mathbf{u}\| = \sqrt{9+1} = \sqrt{10}, \|\mathbf{v}\| = \sqrt{4+1+9} = \sqrt{14}$$

$$\|\mathbf{u}\| \|\mathbf{v}\| = 11.83$$

$$\langle \mathbf{u}, \mathbf{v} \rangle = |-6 - 1 + 0| = 7$$

$$\Rightarrow |\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|$$

⑦ $\mathbf{u} = (-4, 2, 1), \mathbf{v} = (8, -4, -2)$

$$\|\mathbf{u}\| = \sqrt{16+4+1} = \sqrt{21}; \|\mathbf{v}\| = \sqrt{64+16+4} = \sqrt{84} = 2\sqrt{21}$$

$$\|\mathbf{u}\| \|\mathbf{v}\| = 42$$

$$|\langle \mathbf{u}, \mathbf{v} \rangle| = |-32 - 8 - 2|$$

$$= 42$$

Theorem 6.2.3 (Properties of Distance)

- If $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are vectors in an inner product space V , and if k is any scalar, then:
 - $d(\mathbf{u}, \mathbf{v}) \geq 0$
 - $d(\mathbf{u}, \mathbf{v}) = 0$ if and only if $\mathbf{u} = \mathbf{v}$
 - $d(\mathbf{u}, \mathbf{v}) = d(\mathbf{v}, \mathbf{u})$
 - $d(\mathbf{u}, \mathbf{v}) \leq d(\mathbf{u}, \mathbf{w}) + d(\mathbf{w}, \mathbf{v})$ (Triangle inequality)

6-2 Angle Between Vectors

- The Cauchy-Schwarz inequality for R^n (Theorem 4.1.3) follows as a special case of Theorem 6.2.1 by taking $\langle \mathbf{u}, \mathbf{v} \rangle$ to be the Euclidean inner product $\mathbf{u} \cdot \mathbf{v}$.
- The angle between vectors in general inner product spaces can be defined as

$$\cos \theta = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\| \|\mathbf{v}\|} \quad \text{and} \quad 0 \leq \theta \leq \pi$$

- Example 2
 - Let R^4 have the Euclidean inner product. Find the cosine of the angle θ between the vectors $\mathbf{u} = (4, 3, 1, -2)$ and $\mathbf{v} = (-2, 1, 2, 3)$.

Solution

We leave it for the reader to verify that

$$\|\mathbf{u}\| = \sqrt{30}, \quad \|\mathbf{v}\| = \sqrt{18}, \quad \text{and} \quad \langle \mathbf{u}, \mathbf{v} \rangle = -9$$

so that

$$\cos \theta = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\| \|\mathbf{v}\|} = -\frac{9}{\sqrt{30}\sqrt{18}} = -\frac{3}{2\sqrt{15}}$$

Let $\mathbb{R}^2$, $\mathbb{R}^3$, and $\mathbb{R}^4$ have the Euclidean inner product. In each part, find the cosine of the angle between u and v .

5.

(a) $u = (1, -3)$, $v = (2, 4)$

$$u = (1, -3), v = (2, 4)$$

$$\langle u, v \rangle = u_1 v_1 + u_2 v_2 = 2 - 12 = -10. \quad \|u\| = \sqrt{1+9} = \sqrt{10} \quad \|v\| = \sqrt{4+16} = \sqrt{20}$$

$$\cos \theta = \frac{\langle u, v \rangle}{\|u\| \|v\|} = \frac{-10}{\sqrt{10} \cdot \sqrt{20}} = -0.707 \text{ or } -\frac{1}{\sqrt{2}}$$

(b) $u = (-1, 0)$, $v = (3, 8)$

b) $u = (-1, 0)$, $v = (3, 8)$

$$\langle u, v \rangle = -3 \quad \|u\| = 1 \quad \|v\| = \sqrt{9+64} = \sqrt{73}$$

$$\cos \theta = \frac{-3}{\sqrt{73}}$$

(c) $u = (-1, 5, 2), v = (2, 4, -9)$

(d) $u = (4, 1, 8), v = (1, 0, -3)$

(e) $u = (1, 0, 1, 0), v = (-3, -3, -3, -3)$

(f) $u = (2, 1, 7, -1), v = (4, 0, 0, 0)$

① $u = (-1, 5, 2)$

$v = (2, 4, -9)$

$\langle u, v \rangle = -2 + 20 - 18$
 $= 0$

$\cos \theta = \underline{0}$

① $u = (4, 1, 8), v = (1, 0, -3)$

Ans = $\frac{-20}{9\sqrt{10}}$

②

$u = (2, 1, 7, -1)$

$v = (4, 0, 0, 0)$

Ans = $2/\sqrt{55}$

② $u = (1, 0, 1, 0); v = (-3, -3, -3, -3)$

$\langle u, v \rangle = -3 + 0 - 3 + 0 = -6$

$\|u\| = \sqrt{2} \quad \|v\| = \sqrt{9+9+9+9}$
 $= \sqrt{36} = 6$

$\cos \theta = \frac{\langle u, v \rangle}{\|u\| \|v\|} = \frac{-6}{\sqrt{2} \times 6} = \frac{-1}{\sqrt{2}}$

Let $\mathbb{R}^2$, $\mathbb{R}^3$, and $\mathbb{R}^4$ have the Euclidean inner product. In each part, find the cosine of the angle between $\mathbf{u}$ and $\mathbf{v}$.

5.

(a) $\mathbf{u} = (1, -3)$, $\mathbf{v} = (2, 4)$ (e) $\mathbf{u} = (1, 0, 1, 0)$, $\mathbf{v} = (-3, -3, -3, -3)$

(c) $\mathbf{u} = (-1, 5, 2)$, $\mathbf{v} = (2, 4, -9)$

5. (a) $\cos \theta = \frac{\langle (1, -3), (2, 4) \rangle}{\|(1, -3)\| \|(2, 4)\|} = \frac{2 - 12}{\sqrt{10} \sqrt{20}} = \frac{-1}{\sqrt{2}}$

(c) $\cos \theta = \frac{\langle (-1, 5, 2), (2, 4, -9) \rangle}{\|(-1, 5, 2)\| \|(2, 4, -9)\|} = \frac{-2 + 20 - 18}{\sqrt{30} \sqrt{101}} = 0$

(e) $\cos \theta = \frac{\langle (1, 0, 1, 0), (-3, -3, -3, -3) \rangle}{\|(1, 0, 1, 0)\| \|(-3, -3, -3, -3)\|} = \frac{-3 - 3}{\sqrt{2} \sqrt{36}} = \frac{-1}{\sqrt{2}}$

5) $\mathbf{u} = (1, -3)$ $\mathbf{v} = (2, 4)$

$$\cos \theta = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\| \|\mathbf{v}\|} = \frac{2 - 12}{\sqrt{10} \cdot \sqrt{20}} = \frac{-10}{\sqrt{200}} = \frac{-1}{\sqrt{2}}$$

(c) $\mathbf{u} = (-1, 5, 2)$ $\mathbf{v} = (2, 4, -9)$

$$\cos \theta = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\| \|\mathbf{v}\|} = \frac{-2 + 20 - 18}{\sqrt{30} \cdot \sqrt{101}} = 0 = 90^\circ$$

6. Let P_2 have the inner product in Example 8 of Section 6.1. Find the cosine of the angle between p and q .

(a) $p = -1 + 5x + 2x^2$, $q = 2 + 4x - 9x^2$

(b) $p = x - x^2$, $q = 7 + 3x + 3x^2$

(a) $p = -1 + 5x + 2x^2$, $q = 2 + 4x - 9x^2$

$$\langle p, q \rangle = -2 + 20 - 18 = 0$$

$$\cos \theta = 0$$

(b) $p = x - x^2$, $q = 7 + 3x + 3x^2$

$$\langle p, q \rangle = 0 + 3 - 3 = 0$$

$$\cos \theta = 0$$

6: (a) $p = -1 + 5x + 2x^2$, $q = 2 + 4x - 9x^2$

$$\cos \theta = \frac{\langle p, q \rangle}{\|p\| \|q\|} = \frac{-2 + 20 - 18}{\sqrt{30} \sqrt{56}}$$

(b) $p = x - x^2$; $q = 7 + 3x + 3x^2$

$$\cos \theta = \frac{3 - 3}{\sqrt{2} \sqrt{68}} = 0$$

8. Let M_{22} have the inner product in Example 7 of Section 6.1. Find the cosine of the angle between A and B .

(a) $A = \begin{bmatrix} 2 & 6 \\ 1 & -3 \end{bmatrix}, B = \begin{bmatrix} 3 & 2 \\ 1 & 0 \end{bmatrix}$

(b) $A = \begin{bmatrix} 2 & 4 \\ -1 & 3 \end{bmatrix}, B = \begin{bmatrix} -3 & 1 \\ 4 & 2 \end{bmatrix}$

② $A = \begin{bmatrix} 2 & 6 \\ 1 & -3 \end{bmatrix}, B = \begin{bmatrix} 3 & 2 \\ 1 & 0 \end{bmatrix}$

$$\langle A, B \rangle = u_1 v_1 + u_2 v_2 + u_3 v_3 + u_4 v_4$$

$$= 6 + 12 + 1 - 0 = 19.$$

$$\|A\| = \sqrt{4 + 36 + 1 + 9} = \sqrt{50} \quad \|B\| = \sqrt{9 + 4 + 1 + 0} = \sqrt{14}.$$

$$\cos \theta = \frac{\langle A, B \rangle}{\|A\| \|B\|} = \frac{19}{\sqrt{50} \cdot \sqrt{14}} = 0.718 \text{ or } \frac{19}{10\sqrt{7}}.$$

③ $A = \begin{bmatrix} 2 & 4 \\ -1 & 3 \end{bmatrix}, B = \begin{bmatrix} -3 & 1 \\ 4 & 2 \end{bmatrix}$

$$\langle A, B \rangle = -6 + 4 - 4 + 6 = 0.$$

$$\cos \theta = 0.$$

6-2 Orthogonality

Two vectors $\mathbf{u}$ and $\mathbf{v}$ in an inner product space are called **orthogonal** if $\langle \mathbf{u}, \mathbf{v} \rangle = 0$.

• Example 3

- If M_{22} has the inner product defined previously, then the matrices

$$U = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \text{ and } V = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix}$$

are orthogonal, since $\langle U, V \rangle = 1(0) + 0(2) + 1(0) + 1(0) = 0$.

A set of vectors $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ in $\mathbb{R}^n$ is said to be an **orthogonal set** if each pair of distinct vectors from the set is orthogonal, that is, if $\mathbf{u}_i \cdot \mathbf{u}_j = 0$ whenever $i \neq j$.

EXAMPLE 1 Show that $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is an orthogonal set, where

$$\mathbf{u}_1 = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{u}_2 = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}, \quad \mathbf{u}_3 = \begin{bmatrix} -1/2 \\ -2 \\ 7/2 \end{bmatrix}$$

SOLUTION Consider the three possible pairs of distinct vectors, namely, $\{\mathbf{u}_1, \mathbf{u}_2\}$, $\{\mathbf{u}_1, \mathbf{u}_3\}$, and $\{\mathbf{u}_2, \mathbf{u}_3\}$.

$$\mathbf{u}_1 \cdot \mathbf{u}_2 = 3(-1) + 1(2) + 1(1) = 0$$

$$\mathbf{u}_1 \cdot \mathbf{u}_3 = 3\left(-\frac{1}{2}\right) + 1(-2) + 1\left(\frac{7}{2}\right) = 0$$

$$\mathbf{u}_2 \cdot \mathbf{u}_3 = -1\left(-\frac{1}{2}\right) + 2(-2) + 1\left(\frac{7}{2}\right) = 0$$

Each pair of distinct vectors is orthogonal, and so $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is an orthogonal set. See Fig. 1; the three line segments there are mutually perpendicular. ■

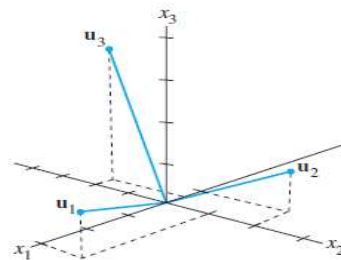


FIGURE 1

In each part, determine whether the given vectors are orthogonal with respect to the Euclidean inner product.

1.

(a) $\mathbf{u} = (-1, 3, 2), \mathbf{v} = (4, 2, -1)$

(b) $\mathbf{u} = (-2, -2, -2), \mathbf{v} = (1, 1, 1)$

(c) $\mathbf{u} = (u_1, u_2, u_3), \mathbf{v} = (0, 0, 0)$

(d) $\mathbf{u} = (-4, 6, -10, 1), \mathbf{v} = (2, 1, -2, 9)$

(e) $\mathbf{u} = (0, 3, -2, 1), \mathbf{v} = (5, 2, -1, 0)$

(f) $\mathbf{u} = (a, b), \mathbf{v} = (-b, a)$

$$\textcircled{a} \quad u = (-1, 3, 2) \quad v = (4, 2, -1)$$

$$\langle u, v \rangle = -4 + 6 - 2 = 0$$

Yes

$$\textcircled{b} \quad u = (-2, -2, -2)$$

$$v = (1, 1, 1)$$

No

$$\textcircled{c} \quad u = (u_1, u_2, u_3), \quad v = (0, 0, 0) \text{ (Yes)}$$

$$\textcircled{d} \quad u = (-4, 6, -10, 1)$$

$$v = (2, 1, -2, 9)$$

No

$$\textcircled{e} \quad u = (0, 3, -2, 1), \quad v = (5, 2, -1, 0) \text{ No}$$

$$\textcircled{f} \quad u = (a, b), \quad v = (-b, a)$$

Yes

10. Let $\mathbb{R}^3$ have the Euclidean inner product. For which values of k are $\mathbf{u}$ and $\mathbf{v}$ orthogonal?

(a) $\mathbf{u} = (2, 1, 3), \mathbf{v} = (1, 7, k)$

(b) $\mathbf{u} = (k, k, 1), \mathbf{v} = (k, 5, 6)$

(a) $\mathbf{u} = (2, 1, 3), \mathbf{v} = (1, 7, k)$
 $\langle \mathbf{u}, \mathbf{v} \rangle = 0 \Rightarrow 2 + 7 + 3k = 0$
 $3k = -9$
 $k = -3$

(b) $\mathbf{u} = (k, k, 1), \mathbf{v} = (k, 5, 6)$
 $\langle \mathbf{u}, \mathbf{v} \rangle = 0 \Rightarrow k^2 + 5k + 6 = 0$
 $k = -2, -3$

2. Do there exist scalars k, l such that the vectors $\mathbf{u} = (2, k, 6)$, $\mathbf{v} = (l, 5, 3)$, and $\mathbf{w} = (1, 2, 3)$ are mutually orthogonal with respect to the Euclidean inner product?

$\mathbf{u} = (2, k, 6), \mathbf{v} = (l, 5, 3), \mathbf{w} = (1, 2, 3)$
 $\langle \mathbf{u}, \mathbf{w} \rangle = 0 \Rightarrow 2 + 2k + 18 = 0 \Rightarrow k = -10$
 $\langle \mathbf{v}, \mathbf{w} \rangle = 0 \Rightarrow l + 10 + 9 = 0 \Rightarrow l = -19$
 $\langle \mathbf{u}, \mathbf{v} \rangle = 0 \Rightarrow 2l + 5k + 18 = 0$

$\therefore$ No scalars k, l do not exist such that all are mutually orthogonal.

3. Let $\mathbb{R}^3$ have the Euclidean inner product. Let $\mathbf{u} = (1, 1, -1)$ and $\mathbf{v} = (6, 7, -15)$. If $\|k\mathbf{u} + \mathbf{v}\| = 13$, what is k ?

3. We have $k\mathbf{u} + \mathbf{v} = (k + 6, k + 7, -k - 15)$, so

$$\begin{aligned}\|k\mathbf{u} + \mathbf{v}\| &= \langle (k\mathbf{u} + \mathbf{v}), (k\mathbf{u} + \mathbf{v}) \rangle^{1/2} \\ &= [(k + 6)^2 + (k + 7)^2 + (-k - 15)^2]^{1/2} \\ &= (3k^2 + 56k + 310)^{1/2}\end{aligned}$$

Since $\|k\mathbf{u} + \mathbf{v}\| = 13$ exactly when $\|k\mathbf{u} + \mathbf{v}\|^2 = 169$, we need to solve the quadratic equation $3k^2 + 56k + 310 = 169$ to find k . Thus, values of k that give $\|k\mathbf{u} + \mathbf{v}\| = 13$ are $k = -3$ or $k = -47/3$.

7. Show that $\mathbf{p} = 1 - x + 2x^2$ and $\mathbf{q} = 2x + x^2$ are orthogonal with respect to the inner product in Exercise 6.

Solution:

$$\langle \mathbf{p}, \mathbf{q} \rangle = 1 - 2 + 2 = 0$$

Yes.

9. Let

$$A = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}$$

Which of the following matrices are orthogonal to A with respect to the inner product in Exercise 8?

(a) $\begin{bmatrix} -3 & 0 \\ 0 & 2 \end{bmatrix}$ 9. (b) $\left\langle \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix} \right\rangle = (2)(1) + (1)(1) + (-1)(0) + (3)(-1) = 0$

(b) $\begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}$

Thus the matrices are orthogonal.

(c) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

(d) $\left\langle \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}, \begin{bmatrix} 2 & 1 \\ 5 & 2 \end{bmatrix} \right\rangle = 4 + 1 - 5 + 6 = 6 \neq 0$

(d) $\begin{bmatrix} 2 & 1 \\ 5 & 2 \end{bmatrix}$

Thus the matrices are not orthogonal.

6-2 Example 4 (Orthogonal Vectors in P_2)

- Let P_2 have the inner product

$$\langle \mathbf{p}, \mathbf{q} \rangle = \int_{-1}^1 p(x)q(x)dx \text{ and let } \mathbf{p} = x \text{ and } \mathbf{q} = x^2.$$

- Then

$$\|\mathbf{p}\| = \langle \mathbf{p}, \mathbf{p} \rangle^{1/2} = \left[\int_{-1}^1 xx dx \right]^{1/2} = \left[\int_{-1}^1 x^2 dx \right]^{1/2} = \sqrt{\frac{2}{3}}$$

$$\|\mathbf{q}\| = \langle \mathbf{q}, \mathbf{q} \rangle^{1/2} = \left[\int_{-1}^1 x^2 x^2 dx \right]^{1/2} = \left[\int_{-1}^1 x^4 dx \right]^{1/2} = \sqrt{\frac{2}{5}}$$

$$\langle \mathbf{p}, \mathbf{q} \rangle = \int_{-1}^1 xx^2 dx = \int_{-1}^1 x^3 dx = 0$$

because $\langle \mathbf{p}, \mathbf{q} \rangle = 0$, the vectors $\mathbf{p} = x$ and $\mathbf{q} = x^2$ are orthogonal relative to the given inner product.